Supplementary Material for

The Blown Lead Paradox: A Pathwise Calibration Benchmark for Win Probability Forecasts

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S1 Proofs of Discrete-Time Results

Throughout, $(p_k)_{k=0}^N$ is a bounded martingale with $p_0 \in (0,1)$ and terminal value $p_N = Y \in \{0,1\}$. For $x \in (0,1]$, define the first-passage time

$$\tau_x := \inf\{k \in \{0,1,\dots,N\} : p_k \geq x\},$$

with $\inf \varnothing := N+1$.

S1.1 Unconditional distribution of M_N

Theorem S1.1 (Discrete-time unconditional bound)**.** Let $M_N = \max_{0 \leq k \leq N} p_k$. For $x \in [p_0, 1)$,

$$\mathbb{P}(M_N < x) \ \geq \ 1 - \frac{p_0}{x},$$

and hence $F_{M_N}(x) = \mathbb{P}(M_N \leq x) \geq 1 - p_0/x$. The strict-threshold bound is an equality if $p_{\tau_x} = x$ a.s. on $\{\tau_x \leq N\}$; equality for $F_{M_N}(x)$ additionally requires $\mathbb{P}(M_N = x) = 0$. For $x < p_0$, $F_{M_N}(x) = 0$. Moreover, M_N has an atom at 1 with $\mathbb{P}(M_N = 1) = p_0$.

Proof. If $x < p_0$ then $M_N \geq p_0 > x$ a.s., hence $F_{M_N}(x) = 0$.

Fix $x \in [p_0, 1)$. By optional stopping for the bounded martingale (p_k) at $\tau_x \wedge N$,

$$\mathbb{E}[p_{\tau_x \wedge N}] = p_0.$$

Decompose $p_{\tau_x \wedge N} = p_{\tau_x}\mathbf{1}\{\tau_x \leq N\} + p_N\mathbf{1}\{\tau_x > N\}$. On $\{\tau_x \leq N\}$ we have $p_{\tau_x} \geq x$. Therefore,

$$p_0 = \mathbb{E}[p_{\tau_x}\mathbf{1}\{\tau_x \leq N\}] + \mathbb{E}[p_N\mathbf{1}\{\tau_x > N\}] \ \geq \ x\,\mathbb{P}(\tau_x \leq N).$$

Since $\{\tau_x \leq N\} = \{M_N \geq x\}$, this yields

$$\mathbb{P}(M_N \geq x) \leq \frac{p_0}{x} \qquad \Rightarrow \qquad \mathbb{P}(M_N < x) \geq 1 - \frac{p_0}{x}.$$

Equality in the strict-threshold bound holds when $p_{\tau_x} = x$ a.s. on $\{\tau_x \leq N\}$. Because $F_{M_N}(x) = \mathbb{P}(M_N < x) + \mathbb{P}(M_N = x)$, the corresponding CDF equality also requires no atom at x.

Finally, $M_N = 1$ iff $p_N = 1$ iff $Y = 1$, so $\mathbb{P}(M_N = 1) = p_0$. □

S1.2 Conditional distribution of M_N given $Y = 0$

Theorem S1.2 (Discrete-time conditional bound)**.** For $x \in [p_0, 1)$,

$$\mathbb{P}(M_N < x \mid Y = 0) \;\geq\; 1 - \Big(\frac{p_0}{1-p_0}\Big)\Big(\frac{1-x}{x}\Big),$$

and hence the same bound holds for $F_{M_N|Y=0}(x) = \mathbb{P}(M_N \leq x \mid Y = 0)$. The strict-threshold bound is an equality under the same no-overshoot condition as in Theorem S1.1; CDF equality additionally requires $\mathbb{P}(M_N = x \mid Y = 0) = 0$. For $x < p_0$, $F_{M_N|Y=0}(x) = 0$.

Proof. Fix $x \in [p_0, 1)$. Decompose $\mathbb{P}(M_N \geq x)$ by Y:

$$\mathbb{P}(M_N \geq x) = \mathbb{P}(M_N \geq x, Y = 1) + \mathbb{P}(M_N \geq x, Y = 0).$$

On $\{Y = 1\}$ we have $p_N = 1 \geq x$, hence $\{Y = 1\} \subseteq \{M_N \geq x\}$ and

$$\mathbb{P}(M_N \geq x) = \mathbb{P}(Y = 1) + \mathbb{P}(Y = 0)\mathbb{P}(M_N \geq x \mid Y = 0) = p_0 + (1 - p_0)\mathbb{P}(M_N \geq x \mid Y = 0).$$

By Theorem S1.1, $\mathbb{P}(M_N \geq x) \leq p_0/x$, so

$$\frac{p_0}{x} \;\geq\; p_0 + (1 - p_0)\mathbb{P}(M_N \geq x \mid Y = 0),$$

which rearranges to

$$\mathbb{P}(M_N \geq x \mid Y = 0) \;\leq\; \Big(\frac{p_0}{1-p_0}\Big)\Big(\frac{1-x}{x}\Big).$$

Taking complements gives the stated strict-threshold bound; the ordinary CDF bound follows because $\mathbb{P}(M_N \leq x \mid Y = 0) \geq \mathbb{P}(M_N < x \mid Y = 0)$. □

Proposition S1.3 (Discrete-time correction identity)**.** For $x \in (p_0, 1)$,

$$\mathbb{P}(M_N \geq x \mid Y = 0) = \frac{p_0}{1-p_0} \cdot \frac{1-x}{x} - \frac{1}{x(1-p_0)}\,\mathbb{E}\big[(p_{\tau_x} - x)\mathbf{1}\{\tau_x < N\}\big] - \frac{1-x}{x(1-p_0)}\,\mathbb{P}(\tau_x = N).$$

Proof. Fix $x \in (p_0, 1)$. Since $x < 1$, the event $\{Y = 1\}$ is contained in $\{\tau_x \leq N\}$, so

$$\mathbb{P}(\tau_x \leq N) = \mathbb{P}(Y = 1) + (1 - p_0)\mathbb{P}(M_N \geq x \mid Y = 0).$$

Optional stopping at $\tau_x \wedge N$ gives

$$p_0 = \mathbb{E}[p_{\tau_x \wedge N}] = \mathbb{E}[p_{\tau_x}\mathbf{1}\{\tau_x \leq N\}] = x\,\mathbb{P}(\tau_x \leq N) + \mathbb{E}\big[(p_{\tau_x} - x)\mathbf{1}\{\tau_x \leq N\}\big].$$

Combining the two displays and rearranging yields

$$\mathbb{P}(M_N \geq x \mid Y = 0) = \frac{p_0}{1-p_0} \cdot \frac{1-x}{x} - \frac{1}{x(1-p_0)}\,\mathbb{E}\big[(p_{\tau_x} - x)\mathbf{1}\{\tau_x \leq N\}\big].$$

Finally,

$$\mathbb{E}\big[(p_{\tau_x} - x)\mathbf{1}\{\tau_x \leq N\}\big] = \mathbb{E}\big[(p_{\tau_x} - x)\mathbf{1}\{\tau_x < N\}\big] + (1 - x)\mathbb{P}(\tau_x = N),$$

because $\tau_x = N$ implies $p_{\tau_x} = p_N = 1$. Substituting gives the stated decomposition. □

S2 Proofs of Continuous-Path Results

Let $(p_t)_{t\in[0,1]}$ be a bounded martingale with $p_0 \in (0,1)$ and $p_1 = Y \in \{0,1\}$, and assume path continuity. Define $M = \sup_{t\in[0,1]} p_t$ and $\tau_x = \inf\{t \in [0,1] : p_t \geq x\}$.

S2.1 Unconditional distribution of M

Theorem S2.1 (Continuous-path unconditional)**.** For $x \in [p_0, 1)$,

$$F_M(x) = \mathbb{P}(M \leq x) = 1 - \frac{p_0}{x},$$

and $F_M(x) = 0$ for $x < p_0$, while $F_M(1) = 1$. In particular, $\mathbb{P}(M = 1) = p_0$.

Proof. If $x < p_0$ then $M \geq p_0 > x$ a.s.

Fix $x \in [p_0, 1)$. Optional stopping at $\tau_x \wedge 1$ yields $\mathbb{E}[p_{\tau_x \wedge 1}] = p_0$. By continuity, on $\{\tau_x < 1\}$ we have $p_{\tau_x} = x$, and on $\{\tau_x \geq 1\}$ we have $p_1 = Y$. Thus

$$p_0 = x\,\mathbb{P}(\tau_x < 1) + \mathbb{E}[Y\mathbf{1}\{\tau_x \geq 1\}].$$

But if $Y = 1$, continuity and $p_1 = 1 > x$ imply that $p_t \geq x$ for some $t < 1$, and hence $\tau_x < 1$; therefore $\mathbb{E}[Y\mathbf{1}\{\tau_x \geq 1\}] = 0$ for $x < 1$. Therefore $p_0 = x\,\mathbb{P}(\tau_x < 1)$, i.e. $\mathbb{P}(M \geq x) = \mathbb{P}(\tau_x < 1) = p_0/x$. Because this tail is continuous in x on $[p_0, 1)$, M has no atoms there. Thus $\mathbb{P}(M \leq x) = \mathbb{P}(M < x) = 1 - p_0/x$ on that interval. Finally, $M = 1$ iff $Y = 1$, so $\mathbb{P}(M = 1) = p_0$. □

S2.2 Conditional distribution of M given $Y = 0$

Theorem S2.2 (Continuous-path conditional)**.** For $x \in [p_0, 1)$,

$$F_{M|Y=0}(x) = \mathbb{P}(M \leq x \mid Y = 0) = 1 - \Big(\frac{p_0}{1-p_0}\Big)\Big(\frac{1-x}{x}\Big),$$

and $F_{M|Y=0}(x) = 0$ for $x < p_0$, while $F_{M|Y=0}(1) = 1$.

Proof. Fix $x \in [p_0, 1)$. As before,

$$\mathbb{P}(M \geq x) = \mathbb{P}(Y = 1) + (1-p_0)\mathbb{P}(M \geq x \mid Y = 0) = p_0 + (1-p_0)\mathbb{P}(M \geq x \mid Y = 0).$$

By Theorem S2.1, $\mathbb{P}(M \geq x) = p_0/x$, hence

$$\frac{p_0}{x} = p_0 + (1-p_0)\mathbb{P}(M \geq x \mid Y = 0) \quad \Rightarrow \quad \mathbb{P}(M \geq x \mid Y = 0) = \Big(\frac{p_0}{1-p_0}\Big)\Big(\frac{1-x}{x}\Big).$$

This conditional tail is continuous in x below one, so it has no atoms there; taking complements yields the stated CDF. Finally, on $\{Y = 0\}$ the process cannot attain 1 (if $p_t = 1$ then $\mathbb{P}(Y = 1 \mid \mathcal{F}_t) = 1$), so $M < 1$ a.s. and $F_{M|Y=0}(1) = 1$. □

S3 Distribution of M_λ (Two Teams)

Let (p_t) and (q_t) be the two teams' win-probability martingales, where $q_t = 1 - p_t$ and $p_0 = \mathbb{P}(Y = 1)$; assume $p_0 \geq 0.5$ without loss of generality. The maximum win probability attained by the eventual loser is

$$M_\lambda \equiv \sup_{0 \leq t \leq 1} \Big(p_t \mathbf{1}\{Y = 0\} + q_t \mathbf{1}\{Y = 1\} \Big). \tag{1}$$

By the law of total probability,

$$F_{M_\lambda}(x) = \mathbb{P}(M_\lambda \leq x) = \mathbb{P}(M_\lambda \leq x \mid Y = 0)(1 - p_0) + \mathbb{P}(M_\lambda \leq x \mid Y = 1)p_0. \tag{2}$$

When team A loses, Theorem S2.2 gives

$$F_{M_\lambda|Y=0}(x) = \mathbb{P}(M_\lambda \leq x \mid Y = 0) = \begin{cases} 0 & \text{for } x < p_0, \\ 1 - \left(\frac{p_0}{1-p_0}\right)\left(\frac{1-x}{x}\right) & \text{for } x \in [p_0, 1), \\ 1 & \text{for } x = 1. \end{cases} \tag{3}$$

Applying the same formula with p_0 replaced by $1 - p_0$ gives

$$F_{M_\lambda|Y=1}(x) = \mathbb{P}(M_\lambda \leq x \mid Y = 1) = \begin{cases} 0 & \text{for } x < 1 - p_0, \\ 1 - \left(\frac{1-p_0}{p_0}\right)\left(\frac{1-x}{x}\right) & \text{for } x \in [1 - p_0, 1), \\ 1 & \text{for } x = 1. \end{cases} \tag{4}$$

With $p_0 \geq 0.5$ (team A favored), substituting (3) and (4) into (2) gives three regions:

1. For $0 \leq x < 1 - p_0$, neither (3) nor (4) contributes, so $F_{M_\lambda}(x) = 0$.
2. For $1 - p_0 \leq x < p_0$, only (4) contributes:

$$\begin{aligned} F_{M_\lambda}(x) &= (1 - p_0) \cdot 0 + p_0 \cdot \left[1 - \left(\frac{1-p_0}{p_0}\right)\left(\frac{1-x}{x}\right)\right] \\ &= p_0 - (1 - p_0)\left(\frac{1-x}{x}\right) = 1 - \frac{1-p_0}{x}. \end{aligned}$$

3. For $x \geq p_0$, both (3) and (4) contribute:

$$\begin{aligned} F_{M_\lambda}(x) &= (1 - p_0)\left[1 - \left(\frac{p_0}{1-p_0}\right)\left(\frac{1-x}{x}\right)\right] + p_0\left[1 - \left(\frac{1-p_0}{p_0}\right)\left(\frac{1-x}{x}\right)\right] \\ &= \left[(1 - p_0) - p_0\left(\frac{1-x}{x}\right)\right] + \left[p_0 - (1 - p_0)\left(\frac{1-x}{x}\right)\right] \\ &= 1 - \frac{1-x}{x} = \frac{2x-1}{x} = 2 - \frac{1}{x}. \end{aligned}$$

Combining the three regions gives the piecewise cumulative distribution function

$$F_{M_\lambda}(x) = \mathbb{P}(M_\lambda \leq x) = \begin{cases} 0 & \text{for } 0 \leq x < 1 - p_0, \\ 1 - \frac{1-p_0}{x} & \text{for } 1 - p_0 \leq x < p_0, \\ 2 - \frac{1}{x} & \text{for } p_0 \leq x < 1, \\ 1 & \text{for } x = 1. \end{cases}$$

S4 Numerical Benchmarks

The piecewise CDF in Supplementary Section S3 gives the selected values in Table S4.1. When the peak threshold satisfies $x \geq p_0$, the tail is universal:

$$\mathbb{P}(M_\lambda \geq x) = \frac{1}{x} - 1.$$

In the intermediate region, $1 - p_0 \leq x < p_0$, the corresponding probability is $(1 - p_0)/x$. The first three rows follow a game with $p_0 = 2/3$ across the boundary between these regions; the remaining rows give the practically important upper-tail thresholds. For discretely reported paths, these continuous-path probabilities are conservative upper bounds.

Table S4.1: Selected benchmark probabilities for the eventual loser's peak.

Case	Starting favorite	Peak threshold x	$\mathbb{P}(M_\lambda \geq x)$
Intermediate region	$p_0 = 2/3$	1/2	2/3
Universal boundary	$p_0 = 2/3$	2/3	1/2
Universal tail	$p_0 = 2/3$	3/4	1/3
Universal tail	$p_0 \leq 0.90$	0.90	1/9 (0.111)
Universal tail	$p_0 \leq 0.95$	0.95	1/19 (0.0526)
Universal tail	$p_0 \leq 0.99$	0.99	1/99 (0.0101)

S5 Dependence Validity and Bootstrap

Intervals and p-values in Sections S5–S7 use $B = 9,999$ season-stratified dyadic draws with seed 20260815.

S5.1 Exchangeable-array specialization

The empirical-process bootstrap theorem of Davezies et al. (2021) and its multiple-observations-per-dyad extension cover zero or repeated sports matches between a pair. The following corollary specializes those results to an equal-game empirical CDF with within-season normalization.

For season s, let n_s be the number of teams, $N_{s,ab}$ the number of games between teams $a < b$, and $U_{s,ab\ell}$ the benchmark score in meeting ℓ. For $h_t(u) = \mathbf{1}\{u < t\}$, define the empirical numerator and denominator

$$\widehat{A}_s(t) = \binom{n_s}{2}^{-1} \sum_{a<b} \sum_{\ell=1}^{N_{s,ab}} h_t(U_{s,ab\ell}), \qquad \widehat{B}_s = \binom{n_s}{2}^{-1} \sum_{a<b} N_{s,ab}.$$

The strict empirical CDF among games in season s is therefore

$$\widehat{F}_s(t-) = \frac{\widehat{A}_s(t)}{\widehat{B}_s}.$$

Its population counterpart is $F_s(t-) = A_s(t)/B_s$, where

$$A_s(t) = \mathbb{E}\left[\sum_{\ell=1}^{N_{s,12}} h_t(U_{s,12\ell})\right], \qquad B_s = \mathbb{E}[N_{s,12}].$$

Let $\omega_{s,n} \geq 0$ be fixed stratification weights, with $\sum_s \omega_{s,n} = 1$ and $\omega_{s,n} \to \omega_s$. The application conditions on the observed season composition and sets these weights to each season's share of analyzed games. Thus

$$\widehat{F}(t-) = \sum_{s=1}^{S} \omega_{s,n} \widehat{F}_s(t-), \qquad F_{\omega,n}(t-) = \sum_{s=1}^{S} \omega_{s,n} F_s(t-).$$

Within season s, let $(W_{s1}, \ldots, W_{sn_s})$ be multinomial team multiplicities from n_s draws. The bootstrap analogues are

$$\begin{aligned}
\widehat{A}_s^*(t) &= \binom{n_s}{2}^{-1} \sum_{a<b} W_{sa} W_{sb} \sum_{\ell=1}^{N_{s,ab}} h_t(U_{s,ab\ell}), \\
\widehat{B}_s^* &= \binom{n_s}{2}^{-1} \sum_{a<b} W_{sa} W_{sb} N_{s,ab}, \qquad \widehat{F}_s^*(t-) = \frac{\widehat{A}_s^*(t)}{\widehat{B}_s^*},
\end{aligned}$$

and $\widehat{F}^*(t-) = \sum_s \omega_{s,n} \widehat{F}_s^*(t-)$. This ratio is exactly the CDF obtained by normalizing the raw game weights $W_{sa} W_{sb}$ within season.

Corollary S5.1 (CDF specialization of the exchangeable-array bootstrap)**.** Let the number of seasons S be fixed, $n_{\min} = \min_s n_s \to \infty$, and $n_{\min}/n_s \to \kappa_s \in (0, 1]$. Within each season, suppose

$$\big(N_{s,ab}, (U_{s,ab\ell})_{\ell \geq 1}\big)_{a<b}$$

is jointly exchangeable and dissociated under team relabeling, and suppose the season arrays are independent. Assume $N_{s,ab} \leq C < \infty$ almost surely and $B_s > 0$. Assume also that the first-order dyadic projection of the normalized CDF process is nondegenerate, meaning that its limiting covariance function is not identically zero. Then, in $\ell^\infty([0, 1])$,

$$\sqrt{n_{\min}}\{\widehat{F}(t-) - F_{\omega,n}(t-)\} \rightsquigarrow \mathbb{G}(t),$$

and, conditionally on the observed arrays,

$$\sqrt{n_{\min}}\{\widehat{F}^*(t-) - \widehat{F}(t-)\} \rightsquigarrow \mathbb{G}(t).$$

Consequently, at continuity points of the limiting supremum law, the conditional distribution of

$$\sqrt{n_{\min}} \sup_t \{\widehat{F}(t-) - \widehat{F}^*(t-)\}$$

consistently estimates the distribution of

$$\sqrt{n_{\min}} \sup_t \{F_{\omega,n}(t-) - \widehat{F}(t-)\}.$$

Proof. The class

$$\mathcal{H}_+ = \{u \mapsto \mathbf{1}(u < t) : t \in [0, 1]\} \cup \{1\}$$

is a pointwise measurable VC class with bounded envelope: rational thresholds form a countable pointwise-dense subclass. Parts 2 and 3 of Supplemental Theorem S1 in Davezies et al. (2021) therefore give joint weak convergence of $\sqrt{n_s}(\widehat{A}_s - A_s, \widehat{B}_s - B_s)$ and its multinomial-team bootstrap counterpart. The bounded cell count supplies the extension's moment conditions. An unordered symmetric dyad can be embedded in the theorem's ordered-array notation by assigning the same cell to (a, b) and (b, a); the common factor of two cancels from the CDF ratio.

Because $B_s > 0$, the map $\Psi(A, B) = A/B$ is Hadamard differentiable, with derivative

$$\dot{\Psi}_{(A_s,B_s)}(\alpha, \beta) = \frac{\alpha}{B_s} - \frac{A_s\beta}{B_s^2}.$$

The bootstrap functional delta method therefore gives the two process limits for each normalized season CDF. The second term carries the denominator fluctuation into the normalized process. A finite independent weighted sum, together with $n_{\min}/n_s \to \kappa_s$, gives the pooled limits. Finally, $f \mapsto \sup_t\{-f(t)\}$ is Lipschitz on $\ell^\infty([0, 1])$, which gives the last assertion directly, without a symmetry argument. □

Corollary S5.2 (Conservative upper-tail test)**.** Suppose in addition that the discrete-feed null satisfies $F_{\omega,n}(t-) \geq t$ for every $t \in [0, 1]$. Let $c^*_{1-\alpha}$ be the conditional $(1 - \alpha)$ quantile of the scaled recentered bootstrap statistic in Corollary S5.1. Define

$$D_n^{\text{upper}} = \sup_t\{t - \widehat{F}(t-)\}$$

and reject when $\sqrt{n_{\min}}D_n^{\text{upper}} > c^*_{1-\alpha}$. This test has asymptotic size at most α whenever the limiting supremum distribution is continuous at its $(1 - \alpha)$ quantile.

Proof. Under the null,

$$t - \widehat{F}(t-) \leq F_{\omega,n}(t-) - \widehat{F}(t-)$$

pointwise. The result follows by taking suprema and applying Corollary S5.1. □

Corollaries S5.1–S5.2 use a dense-array asymptotic and a nondegenerate first-order dyadic projection. A vanishing projection can produce a second-order limit and conservative pigeonhole resampling; more generally, Menzel (2021) rules out uniform bootstrap consistency across all multiway-dependence regimes. Professional schedules have fixed conference and division structures, and franchises persist across seasons. Section S8 evaluates finite-sample size on the observed schedules, and the franchise-linked specification in Table S7.3 carries team effects across seasons.

S5.2 Season-stratified dyadic pigeonhole bootstrap

Write game g in season s as the dyad (a_g, b_g). The resampling algorithm is:

1. Within each season, sample its observed teams multinomially with replacement and record team multiplicities W_{sa}.

2. Give game g raw weight $W_{sa_g}W_{sb_g}$. Repeat meetings between the same teams therefore receive the same endpoint weight. If a season draw assigns zero total weight to its observed schedule, redraw that season.

3. Normalize the positive game weights within season to sum to that season's observed number of contributions. This retains the empirical season composition while allowing arbitrary dependence among games that share a team.

4. Reuse the same team multiplicities for the PIT process, tail rates, first-crossing residuals, and fixed-clock calibration cells.

This applies Owen's pigeonhole principle—resampling indexing units rather than treating cells as independent—to a symmetric dyadic schedule (Owen, 2007). The exchangeable-array empirical-process results of Davezies et al. (2021) cover recentered empirical CDF and Kolmogorov–Smirnov processes for dyads, including repeated observations between the same pair.

For tied PIT scores $u_1, \ldots, u_n$, the observed upper statistic is computed exactly as

$$D_n^{\text{upper}} = \max_{u \in \{u_1, \ldots, u_n\}} \{u - \hat{F}_U(u-)\}.$$

For bootstrap draw b, let $\hat{F}^*_{U,b}$ be the season-normalized weighted CDF and calculate

$$D^*_{n,b} = \sup_t \{\hat{F}_U(t) - \hat{F}^*_{U,b}(t)\}.$$

For the common observed support, this right-continuous supremum equals the strict-CDF supremum in Corollary S5.1: both empirical CDFs have the same total mass, so their right-limit and left-limit differences run through the same values, with zero at opposite endpoints. The finite-bootstrap p-value is $(1 + \sum_b \mathbf{1}\{D^*_{n,b} \geq D_n^{\text{upper}}\})/(B+1)$.

S6 Published Feeds and Fixed-Time Calibration

Both applications contain completed regular-season games only. NFL seasons are labeled by starting year and include January games from the following calendar year; NBA seasons are labeled by ending year. Because ESPN's NBA season-type field also includes All-Star exhibitions, both participants must be NBA franchises; this excludes 10 All-Star or Rising Stars events, including 5 with usable probability paths. Table S6.1 gives the complete acquisition ledger. "Eligible" means completed and non-tied, while "analyzed" additionally requires a usable public probability path.

Each application retains its published resolution. The NFL paths contain 325,922 updates, with median 175 per game (interquartile range 167–185); the NBA paths contain 3,877,820, with median 466 (447–488).

Table S6.1: Regular-season schedule and public-feed coverage by season.

NFL

Season	Scheduled	Eligible	Analyzed	Missing
2018	256	254	254	0
2019	256	255	255	0
2020	256	255	255	0
2021	272	271	271	0
2022	272	269	269	0
2023	272	272	272	0
2024	272	272	272	0

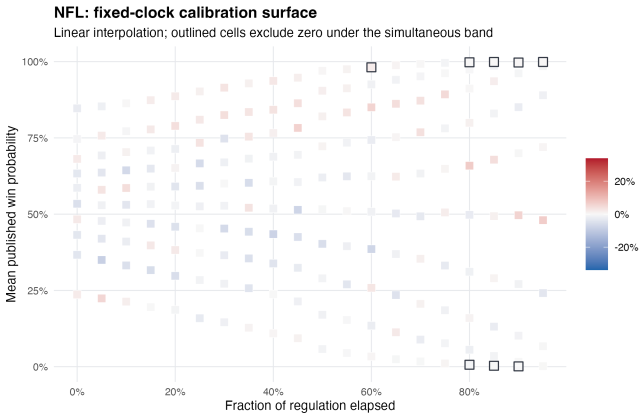
NBA

Season	Scheduled	Eligible	Analyzed	Missing
2018	1,230	1,230	1,222	8
2019	1,230	1,230	1,230	0
2020	1,062	1,059	1,053	6
2021	1,112	1,080	1,080	0
2022	1,241	1,230	1,230	0
2023	1,231	1,230	1,230	0
2024	1,233	1,231	1,231	0

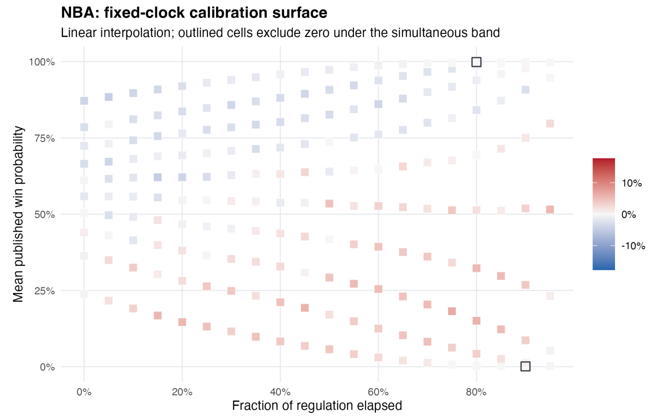
S6.1 Yeh-style fixed-clock comparison

Following the calibration-surface motivation of Yeh et al. (2022), every retained home-team forecast is evaluated at $t = 0, .05, \ldots, .95$ of regulation. Linear interpolation is primary and last observation carried forward (LOCF) is the sensitivity. At each time and for each method, empirical probability quantiles define at most ten bins; a repeated forecast value is never split between bins. Every complete game therefore contributes exactly once to every time point. Overtime updates are omitted from this common-clock comparison but remain in the collapse statistic.

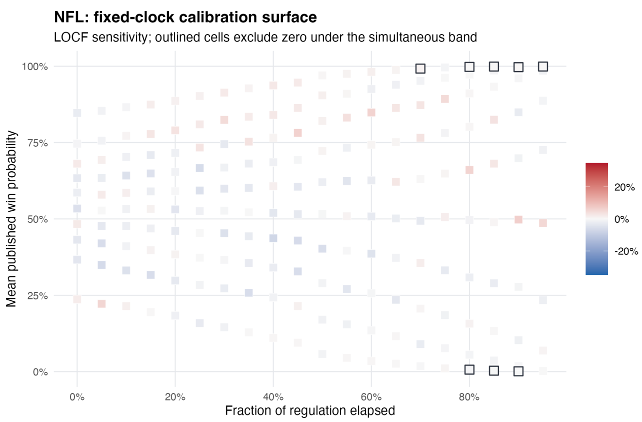
The linear and LOCF analyses use the same 1,848 NFL games and 8,265 NBA games. All NFL games have a complete common clock; 11 NBA paths are incomplete by $t = .95$ and are excluded from this comparison only. For each time–probability cell, the observed-minus-forecast gap is resampled using the common dyadic draws. A simultaneous band is obtained from the 95th percentile of the maximum absolute studentized gap over all cells. Under linear interpolation, the simultaneous bands exclude zero in 8 of 197 NFL cells and 2 of 196 NBA cells; LOCF gives 8 of 197 and 3 of 195, respectively. The flagged cells lie almost entirely at or after 80% of regulation and in the extreme probability bins. In each flagged cell, the observed win rate is exactly zero or one while the corresponding mean forecast remains just inside the boundary. Thus the fixed-clock analysis finds sparse late-game boundary discrepancies, while the interior cells remain within the simultaneous bands. The descriptive root-mean-square cell gaps are 2.9% (linear) and 2.6% (LOCF) in the NFL, and 2.7% and 2.8% in the NBA. Figure S6.1 displays the full surfaces.



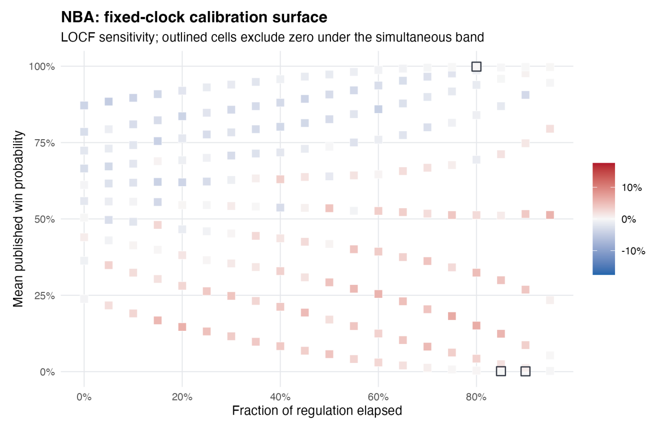
(a) NFL, linear interpolation



(b) NBA, linear interpolation



(c) NFL, LOCF



(d) NBA, LOCF

Figure S6.1: Fixed-clock calibration surfaces. Color is observed win rate minus mean published probability; outlined cells exclude zero under the simultaneous dyadic band. The bin locations adapt to the forecast distribution at each time.

The fixed-clock surfaces evaluate reliability at prespecified times; the collapse benchmark evaluates the selected losing path.

S7 Full Collapse Results and Robustness

S7.1 Full tail and first-crossing results

Table S7.1 reports upper-tail rates at 0.90, 0.95, and 0.99. The NBA excess is positive at 0.90 and 0.95; its 0.99 interval includes zero.

For a team-side path, let τ_q be the first published update at or above q with positive game time remaining. If no preterminal crossing occurs, set $\tau_q = N$ and retain the indicator of crossing. Because τ_q is bounded and $\{\tau_q < N\} \in \mathcal{F}_{\tau_q}$, sequential calibration implies

$$\mathbb{E}\{(p_{\tau_q} - Y)\mathbf{1}(\tau_q < N)\} = 0.$$

Optional stopping therefore centers the first-crossing residual at zero under sequential calibration. Table S7.2 reports $\overline{1-Y}$, $\overline{1-p_{\tau_q}}$, and their difference $\bar{R} = \overline{p_{\tau_q} - Y}$. The estimand is an average

Table S7.1: Upper-tail rates and excess over the continuous benchmark. Intervals are two-sided percentile dyadic-bootstrap intervals for the excess.

League	Threshold	Tail rate	Benchmark rate	Excess (95% CI)
NFL	0.90	8.5%	10.0%	-1.5% [-3.5%, 0.7%]
NFL	0.95	4.5%	5.0%	-0.5% [-2.0%, 1.1%]
NFL	0.99	1.5%	1.0%	0.5% [-0.3%, 1.5%]
NBA	0.90	12.7%	10.0%	2.7% [1.4%, 4.0%]
NBA	0.95	6.3%	5.0%	1.3% [0.4%, 2.3%]
NBA	0.99	1.1%	1.0%	0.1% [-0.3%, 0.5%]

over crossing episodes; when two episodes come from the same game, they receive the same dyadic game weight. The crossing estimand uses a published probability $p \geq q$; the tail estimand in Table S7.1 uses the game-level benchmark score $U \geq q$.

Table S7.2: Preterminal first crossings. The interval is for the mean residual $\bar{R}$.

League	q	Episodes	Games	Observed loss	Implied loss	$\bar{R}$ (95% CI)
NFL	0.90	1,974	1,818	8.5%	7.6%	0.9% [-0.9%, 2.8%]
NFL	0.95	1,842	1,772	4.6%	3.3%	1.3% [-0.2%, 2.9%]
NFL	0.99	1,705	1,686	1.6%	0.5%	1.1% [0.2%, 2.2%]
NBA	0.90	9,371	8,250	12.5%	8.8%	3.7% [2.7%, 4.7%]
NBA	0.95	8,663	8,156	6.5%	4.2%	2.3% [1.4%, 3.2%]
NBA	0.99	7,660	7,598	1.3%	0.8%	0.5% [0.1%, 0.9%]

S7.2 Feed and specification sensitivity

Table S7.3 reports sensitivity to feed corrections, rounding, dependence structure, threshold convention, and time sampling. The crossing rows report the 0.95 residual, while the fixed-clock rows report descriptive root-mean-square calibration gaps over time–probability cells. For reported probabilities rounded to three decimals, lower and upper PIT vectors are obtained by allowing each probability to range within ± 0.0005 and using monotonicity of the benchmark transform. The dyad-plus-calendar row multiplies the team multiplicity by a circular two-week block multiplicity within each season, retaining team dependence while allowing common model-period shocks. The franchise-linked row maps relocations to a common franchise and applies one team multiplicity across all seasons, allowing persistent team effects.

At 0.95, strict versus weak tail counting differs only through the observed atom: the weak and strict NBA rates are 6.3% and 6.3%, with 0.00% exactly at the boundary; the corresponding NFL values are 4.5%, 4.5%, and 0.00%. The path audit identified 16 NBA and 4 NFL games containing loser-side exact-one feed values. The raw, corrected, and exclusion rows in Table S7.3 use identical team draws.

S7.3 Season persistence

The acquisition ledger in Table S6.1 gives explicit missing-feed accounting for both leagues. NFL coverage is complete; minimum NBA season coverage is 99.3%.

Table S7.3: Feed and specification sensitivity. Brackets denote 95% intervals where shown.

Specification	NFL	NBA
Global, corrected: D (p)	0.015 (0.736)	0.097 ($<$ 0.001)
Global, raw: D (p)	0.015 (0.735)	0.097 ($<$ 0.001)
Exclude patched games: D (p)	0.014 (0.769)	0.096 ($<$ 0.001)
Rounding bound for D	[0.013, 0.016]	[0.095, 0.099]
Rounding bound for $\widehat{\mathbb{P}}(U \geq .95)$	[4.5%, 4.5%]	[6.3%, 6.6%]
Dyad plus calendar: D (p)	0.015 (0.856)	0.097 ($<$ 0.001)
Franchise linked: D (p)	0.015 (0.705)	0.097 ($<$ 0.001)
Crossing, preterminal $p \geq .95$	1.3% [-0.2%, 2.9%]	2.3% [1.4%, 3.2%]
Crossing, preterminal $p > .95$	1.3% [-0.2%, 2.9%]	2.3% [1.5%, 3.2%]
Crossing, terminal-inclusive	1.2% [-0.2%, 2.7%]	2.3% [1.4%, 3.1%]
Fixed clock, linear RMS	2.9%	2.7%
Fixed clock, LOCF RMS	2.6%	2.8%

Table S7.4 gives descriptive stability summaries after omitting one season at a time. Across omissions, the NBA global statistic ranges from 0.093 to 0.102, while the NFL statistic ranges from 0.008 to 0.025.

Table S7.4: Leave-one-season-out global statistic and 0.95 tail excess.

NFL

Omitted	Games	D	Excess
2018	1,594	0.017	-0.7%
2019	1,593	0.013	-0.3%
2020	1,593	0.014	-1.0%
2021	1,577	0.013	-0.3%
2022	1,579	0.008	-0.8%
2023	1,576	0.024	-0.4%
2024	1,576	0.025	0.0%

NBA

Omitted	Games	D	Excess
2018	7,054	0.093	1.5%
2019	7,046	0.097	1.2%
2020	7,223	0.094	1.3%
2021	7,196	0.100	1.3%
2022	7,046	0.102	1.3%
2023	7,046	0.093	1.2%
2024	7,045	0.101	1.5%

Across specifications, the NBA departure remains a recurrent upper-tail feature. The tail excess is positive at 0.90 and 0.95, first-crossing residuals are positive at all three thresholds, and the global test rejects for the raw paths, corrected paths, patched-game exclusion, calendar blocks, and franchise linkage. Across the leave-one-season-out samples, D_n^{upper} ranges from 0.093 to 0.102 and the 0.95 tail excess ranges from 1.2% to 1.5%. The game-level 0.99 tail interval includes zero, marking the limit of the upper-tail evidence. Across specifications, the NFL retains the global conclusion of no detected departure; its positive 0.99 first-crossing residual is a localized result.

S8 Finite-Schedule Null Simulation

Finite-sample Type I error is evaluated while retaining the teams and matchups in each observed season schedule. For game g between teams a_g and b_g, generate

$$Z_g = \sqrt{\rho}\,(A_{s,a_g} + A_{s,b_g})/\sqrt{2} + \sqrt{1-\rho}\,E_g, \qquad U_g = \Phi(Z_g),$$

where team effects and game shocks are independent standard normals and team effects are redrawn by season. Marginally, Z_g is standard normal and U_g is uniform, placing every simulation on the boundary of the continuous calibration null. Games sharing a team are correlated through the team effect, with ρ controlling the strength of that dependence. For each $\rho \in \{0, 0.25, 0.5\}$, the global test is applied to 250 Monte Carlo samples using 399 inner pigeonhole draws. The proportion rejected estimates finite-schedule Type I error at nominal level 5%. When $\rho = 0$, the pigeonhole procedure is expected to be conservative because the dyadic projection is degenerate; this is the regime in which multiway bootstrap behavior requires particular care (Menzel, 2021).

Table S8.1: Empirical 5% rejection rates in the finite-schedule null simulation (250 Monte Carlo samples; 399 inner draws).

League	ρ	Dyadic rejection rate
NFL	0	0.0%
NFL	0.25	0.8%
NFL	0.5	3.2%
NBA	0	0.0%
NBA	0.25	3.2%
NBA	0.5	4.4%

The test is conservative under these schedule-specific nulls. Rejection rates rise with the shared-team component, from 0.0% to 3.2% on the NFL schedule and from 0.0% to 4.4% on the NBA schedule, remaining at or below the nominal 5% level. The simulation targets finite-schedule size under shared-team dependence; calendar and cross-season dependence are examined through the sensitivities in Section S7.

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